CSE 202: Object Oriented Programming — Unit 2

Pointers, Arrays & Strings Practice Sheet (Shuffled Order)

1. What will be the output of the following code?

```
int main() {
    int arr[] = {10, 20, 30};
    int *p = arr;
    cout << *(p + 1);
    return 0;
}
```

A. 10

B. 20

C. 30

D. Compilation Error

Correct Answer: B. 20
Explanation: `p` points to `arr[0]` (10). Pointer arithmetic `p + 1` moves the pointer to `arr[1]`, so dereferencing `*(p + 1)` yields `20`.

2. What will be the output of the following code?

```
int main() {
    string s1 = "Hello";
    string s2 = s1;
    s2[0] = 'J';
    cout << s1[0] << " " << s2[0];
    return 0;
}
```

A. H J

B. J J

C. H H

D. Compilation Error

> **Correct Answer:** A. H J
> *Explanation:* `std::string` in C++ performs a deep copy upon assignment. Modifying `s2[0]` to `'J'` does not alter `s1`, so `s1[0]` remains `'H'` and `s2[0]` becomes `'J'`.

3. What will be the output of the following code?

```
int main() {
    int *ptr = NULL;
    cout << *ptr;
    return 0;
}
```

A. 0

B. Garbage value

C. Compilation Error

D. Runtime Error (Segmentation Fault)

> **Correct Answer:** D. Runtime Error (Segmentation Fault)

4. What will be the output of the following code?

```
int main() {
    int x = 10, y = 20;
    int *p = &x;
    int *q = &y;
    p = q;
    cout << *p;
    return 0;
}
```

A. 10

B. 20

C. 30

D. Compilation Error

Correct Answer: B. 20
Explanation: `p` is reassigned to point to the address held by `q` (&y). Dereferencing `*p` outputs the value of `y`, which is `20`.

5. What will be the output of the following code?

```
int main() {
    int x = 5;
    int *p = &x;
    int **pp = &p;
    cout << **pp;
    return 0;
}
```

A. 5

B. Address of p

C. Compilation Error

D. Runtime Error

Correct Answer: A. 5
Explanation: `pp` is a pointer-to-pointer pointing to `p`. Dereferencing twice (`**pp`) yields the value stored in variable `x`, which is `5`.

6. What will be the output of the following code?

```
int main() {
    char *str = "Hello";
    cout << str[1];
    return 0;
}
```

A. e

B. H

C. l

D. Compilation Error

Correct Answer: A. e
Explanation: `str[1]` accesses index 1 of the string literal `"Hello"`, which is `'e'`.

7. What will be the output of the following code?

```
int main() {
    void *ptr = malloc(sizeof(int));
    int *p = (int*)ptr;
    *p = 100;
    cout << *p;
    free(ptr);
    return 0;
}
```

A. 100

B. Garbage value

C. 0

D. Compilation Error

Correct Answer: A. 100

8. What will be the output of the following code?

```
int main() {
    int arr[2][3] = {{1, 2, 3}, {4, 5, 6}};
    cout << *(arr[0] + 2);
    return 0;
}
```

A. 1

B. 2

C. 3

D. 4

Correct Answer: C. 3
Explanation: `arr[0]` is a pointer to the first row `{1, 2, 3}`. `arr[0] + 2` shifts the pointer to index 2 (`3`). Dereferencing `*(arr[0] + 2)` gives `3`.

9. What will be the output of the following code?

```
int main() {
    string str = "Hello";
    cout << str[0] << " " << str.length();
    return 0;
}
```

A. H 5

B. Hello 5

C. H 4

D. Compilation Error

Correct Answer: A. H 5

10. What will be the output of the following code?

```
int main() {
    int arr[3] = {1, 2, 3};
    int *p = arr;
    p++;
    cout << *p;
    return 0;
}
```

A. 1

B. 2

C. 3

D. Compilation Error

Correct Answer: B. 2
Explanation: `p` initially points to `arr[0]` (1). Incrementing `p++` moves it to `arr[1]`, so `*p` outputs `2`.